

TRUSTSCAN 360

Explainable Digital Trust & Safety Platform

Detect • Protect • Verify • Trace • Secure • Build Trust

Technical Documentation & Architecture Specification
Hackathon Comprehensive Submission Document
Implementation Version: 2.0.0 (Production Verified)
Verified Against Actual Codebase Architecture

Executive Overview: TrustScan 360 is a digital trust platform engineered to evaluate whether online content, messages, URLs, documents, credentials, and claims can be trusted. Built upon a unified Six-Pillar architecture, the platform pairs high-speed multimodal AI reasoning (Gemini 3.8 Flash) with deterministic cybersecurity telemetry (VirusTotal 92-engine intelligence, Error Level Analysis, OpenCV matrix decoding, Ed25519 digital signatures, and SHA-256 canonical privacy commitments). Every assessment answers four human questions: *Is this real? Is this dangerous? Why did you reach this conclusion? What should I do next?*

Notice: Detect • Protect • Verify • Trace • Secure • Build Trust

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1. DOCUMENT PURPOSE

This document provides a comprehensive, rigorous technical specification of **TRUSTSCAN 360**. It covers:

- The exact threat landscape addressed by digital trust engineering.
- Every implemented frontend and backend functionality.
- Underlying mathematical foundations, algorithms, and cryptographic implementations.
- Multi-modal AI pipelines (Google Gemini 3.8 Flash, VirusTotal v3 API, OpenCV, Error Level Analysis, Ed25519 digital signatures, difference perceptual hashing).
- Architectural workflows, data flows, security boundaries, and privacy mechanisms.
- Clear distinctions between **IMPLEMENTED**, **PARTIALLY IMPLEMENTED**, and **FUTURE WORK / PLANNED** components.

This documentation serves technical mentors, hackathon evaluators, cybersecurity researchers, and software engineers seeking to inspect, run, or extend the codebase.

2. EXECUTIVE SUMMARY

What is TrustScan 360?

TrustScan 360 is a digital trust and safety web application designed to help users evaluate whether digital content, messages, URLs, documents, credentials, claims, and media can be trusted.

In modern communication channels (WhatsApp, SMS, Telegram, web pages, PDFs), users are continuously subjected to:

- Coercive social engineering and fake bank account suspension notices.
- Newly registered phishing domains hosted on free subdomains.
- Malicious QR codes weaponized for UPI payment fund extraction.

- Forged identity handles, impersonation vectors, and telecom SIM-swap hijackings.
- Manipulated screenshots, fake payment confirmation slips, and synthetic media.

Rather than acting as an opaque "black-box" classifier that merely outputs "SCAM" or "LEGIT", TrustScan 360 operates on a foundational principle:

Notice: Trust isn't a guess. It's evidence.

The Core Operational Flow:



The system answers four fundamental human questions:

1. **Is this real or fake?**
2. **Is this dangerous or suspicious?**
3. **Why did you reach this conclusion?**
4. **What should I do next?**

3. PROBLEM STATEMENT

The Crisis of Unverified Digital Content

The modern digital ecosystem suffers from severe verification asymmetry:

1. **Deceptive Phishing Subdomains:** Cybercriminals deploy lookalike phishing portals on free hosting providers (`*.vercel.app`, `*.pages.dev`, `*.firebaseapp.com`, `*.github.io`) that bypass traditional static URL filters because the parent domains enjoy high search-engine reputation.
2. **Coercive Urgency & Social Engineering:** Fraudsters use psychological intimidation ("Your electricity line will be disconnected within 2 hours", "Your PAN card is blocked", "SBI account suspended") to force hasty credential entry.
3. **UPI Payment Collection Exploits:** In peer-to-peer payment ecosystems (such as Indian UPI), scammers trick victims by sending QR codes or payment collect requests claiming the victim will "receive a prize" if they scan and enter their MPIN. In truth, an MPIN only authorizes debiting funds.
4. **Synthetic Generation & Compression Splicing:** Visual manipulation tools make it effortless to forge banking transaction receipts, alter certificate dates, or fabricate official government letters.
5. **Opaque "Black-Box" AI Classifiers:** Conventional AI filters provide ungrounded binary labels without citations, evidence, or actionable guidance, leaving users helpless when facing false positives or sophisticated threats.

4. HACKATHON REQUIREMENT MAPPING

The following matrix documents the Six-Pillar alignment of TrustScan 360:

Pillar	Requirement	Implemented Feature in TrustScan 360	How It Addresses It	Implementation Status
DETECT	Media & Content Manipulation	Forensics Engine (ELA, EXIF, Copy-Move) + Gemini Multimodal Vision + SynthID frequency inspection	Inspects image compression artifacts, EXIF editing tags, and generative diffusion signatures.	IMPLEMENTED
PROTECT	Identity & Cellular Impersonation	SIM-Swap Risk Self-Assessment + Channel Sender Audit + HIBP Password Prefix API	Analyzes user-reported cellular symptoms against hijacking kill-chains; audits sender handle consistency.	IMPLEMENTED
VERIFY	Documents, Exams & Credentials	PDF Structural Forensics + Ed25519 Exam Integrity Seal + W3C Verifiable Credentials	Verifies structural PDF object trees, validates asymmetric signatures, and evaluates visual tampering via dHash.	IMPLEMENTED
TRACE	Claims & Visual Provenance	Atomic Claim Extraction (Gemini) + Google Fact Check Tools API / IFCN Telemetry + Provenance Timeline	Decomposes viral text into testable atomic claims and matches against published IFCN journalistic debunks.	IMPLEMENTED
SECURE	URLs, Scams & Phishing	10-Step Safe URL Crawler + VirusTotal 92-Engine Telemetry + QR Matrix Decoder (OpenCV)	Safely fetches DOM structure, flags sensitive input fields, audits domain vs brand, and detects UPI traps.	IMPLEMENTED
BUILD TRUST	Explainability & Verifiability	Standardized Final Trust Report + "Show Me Why" Exact Evidence + SHA-256 Trust Passport + /verify/{id}	Delivers plain-English explanations, exact evidence cards, and public tamper-proof audit passports.	IMPLEMENTED

5. COMPLETE FEATURE LIST

Catalog of Implemented Functionalities:

- SMS / Phishing Text Scanner:** NLP semantic fraud and urgency detector.
- Screenshot / Payment Receipt Scanner:** Multimodal visual forensics and OCR tampering analysis.
- 10-Step URL Web-Safety Investigation Pipeline:** Safe HTTP crawler, DOM form auditor, VirusTotal aggregator.
- QR Code Matrix & Payment Auditor:** Computer-vision matrix decoder with UPI debit trap protection.
- Claim Trace & Fact-Check Engine:** Atomic statement decomposition and fact-checking registry search.
- PDF Document Forensic Inspector:** Structural object tree parsing and incremental update audit.

7. **Exam Integrity Seal & Perceptual Hash Verification:** Ed25519 asymmetric signature and 64-bit gradient dHash.
8. **W3C Verifiable Credential Issuer & Validator:** Cryptographic JSON-LD credential signature checker.
9. **Deepfake Visual & Frequency Analyzer:** Error Level Analysis (ELA) heatmap and diffusion anomaly detection.
10. **SIM-Swap & Telecom Hijack Risk Evaluator:** Multi-symptom cellular integrity diagnostic checklist.
11. **HavelBeenPwned k-Anonymity Breach Checker:** SHA-1 5-character prefix credential safety lookup.
12. **Attack Chain Correlator:** Kill-chain timeline mapper connecting reconnaissance, delivery, and exploitation.
13. **Verifiable Trust Passport Engine:** SHA-256 canonical metadata commitment and public `/verify/{id}` page.
14. **Zero-Knowledge Ready Privacy Layer:** Blinded commitment engine strictly keeping original inputs private.
15. **Transparent Reliability Metrics:** System-level transparency score calculating grounded evidence density.

6. TEXT SCANNER

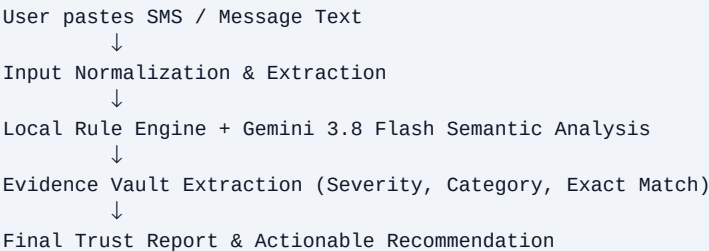
What It Does

Inspects text messages, SMS alerts, WhatsApp forwards, or email snippets for deceptive manipulation, urgent threats, impersonation, and fraudulent payment prompts.

Why It Exists

Attackers frequently disguise phishing vectors as urgent institutional notifications (e.g. State Bank of India, Income Tax Department, FedEx, electricity boards). Users need instant clarity on whether a text message is legitimate or social engineering.

User Workflow



How It Works Technically

1. **Client Ingestion:** Receives raw text from `#input-tool-text` via **POST /api/case** or direct text assessment.
2. **Deterministic Pattern Rules:**
 - Detects UPI collect request language: **collect request, enter pin to receive, approve request.**
 - Detects extortion/harassment keywords: **contact all your relatives, legal notice, police coming.**
 - Detects known deceptive URL patterns: **bit.ly, tinyurl, ngrok, login-sbi, kyc-update.**
3. **Gemini 3.8 Flash Analysis (gemini_service.py):**
 - Dispatches structured prompt with **response_mime_type="application/json"**.
 - Evaluates: urgency pressure, impersonation signals, unverified banking claims.

4. **Fail-Safe Cognitive Fallback:** If API limits are reached, executes local heuristic evaluation flagging coercive terms (**urgent, kyc, block, suspend, penalty, pan, sbi**).

Technologies Used

- Python 3.12, FastAPI, Pydantic 2.x
- Google Gemini API SDK (**google-genai / gemini-3.8-flash**)
- Regular Expressions (**re**) for URL and keyword pattern extraction

Output Example

```
{
  "risk_level": "DANGEROUS",
  "overall_risk_score": 88,
  "headline": "HIGH-RISK IMPERSONATION & KYC SCAM DETECTED",
  "plain_english_explanation": "This message uses false urgency claiming your bank account is suspended to coerce you into clicking an unauthorized update link.",
  "evidence_vault": [
    {
      "source": "Rules",
      "category": "social_engineering",
      "severity": "HIGH",
      "title": "Psychological Pressure Tactics Detected",
      "description": "Message exhibits coercive artificial urgency threatening account blockage."
    }
  ]
}
```

Limitations

Cannot verify offline private arrangements between parties; relies on semantic and linguistic indicators.

7. SCREENSHOT / IMAGE FORENSICS ANALYSIS

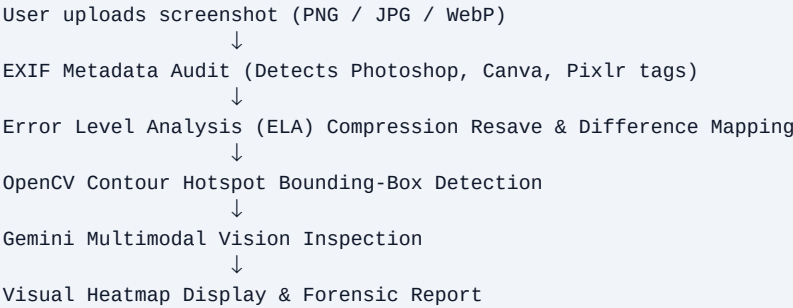
What It Does

Audits uploaded screenshots (payment confirmations, banking alerts, WhatsApp chats) for image compression anomalies, altered text blocks, and editing metadata.

Why It Exists

Fraudulent payment screenshot generators allow scammers to create fake Google Pay, PhonePe, or bank transfer receipts to deceive shopkeepers or individuals.

User Workflow



How It Works Technically

- 1. EXIF Inspection (forensics_service.py):** Extracts camera/device tags. Flags editing software tags (**Photoshop, GIMP, Canva, Pixlr, Procreate**).
- 2. Error Level Analysis (ELA):**
 - Re-saves the image at a controlled JPEG quality factor ($Q = 90$).
 - Computes pixel-wise difference: $\Delta(x, y) = |I_{\text{orig}}(x, y) - I_{\text{resaved}}(x, y)|$.
 - Stretches difference brightness across dynamic range: $Scale = \frac{255}{\max(\Delta)}$.
 - Converts to grayscale, applies binary thresholding at value 180, and runs **cv2.findContours** to locate spliced bounding boxes.
- 3. Multimodal Vision:** Dispatches raw image bytes to Gemini Vision to detect semantic inconsistencies in receipt fonts and transaction layouts.

Technologies Used

- Pillow (**PIL . Image, PIL . ImageChops, PIL . ImageEnhance**)
- OpenCV (**cv2**) for contour analysis and bounding-box geometry
- NumPy (**numpy**) array manipulation

Output Example

- Generates an interactive visual heatmap rendered in base64 on **#report-ela-image**.
- Flags detected compression hotspots with coordinates $[y_{\min}, x_{\min}, y_{\max}, x_{\max}]$.

Limitations

Multiple repeated resaves or messaging app compression (e.g. WhatsApp) naturally alter high-frequency compression data, which the system explicitly flags as a disclaimer.

8. URL ANALYZER & 10-STEP INVESTIGATION PIPELINE

What It Does

Investigates suspicious links, domain names, and web addresses across an ordered 10-step cybersecurity investigation workflow, evaluating live DOM structure and 92 VirusTotal security engines.

Why It Exists

Modern phishing sites evade detection by using free subdomains (`*.vercel.app`, `*.pages.dev`, `*.github.io`) that have 0 VirusTotal hits on day one. A URL scanner must safely inspect the page's actual contents and forms without putting the user at risk.

User Workflow

```

User submits suspicious URL
    ↓
Safe HTTP Crawler (No client JS execution, no form submit)
    ↓
DOM Parsing: Title, Redirects, Form Inputs, Brand Clues, Urgency Keywords
    ↓
Parallel Execution:
  ■ VirusTotal v3 Threat Intelligence (92 Security Vendors)
  ■ Gemini 10-Step Web-Safety Pipeline Prompt
    ↓
Local Cognitive Fail-Safe Fallback Verification
    ↓
Render 10-Step Interactive Cards on Front-End
  
```

The 10 Steps in Order:

1. **FETCH THE PAGE:** Safely reads page title, status code, and final redirected destination.
2. **WHAT IT CLAIMS TO BE:** Identifies the claimed corporate brand or institution from logo text, titles, or copyright notices.
3. **WHERE IT ACTUALLY LIVES:** Inspects host domain vs claimed brand. Flags free hosting subdomains, odd TLDs (`.xyz`, `.top`, `.online`), and typosquatting.
4. **WHAT IT ASKS FOR:** Audits form `<input>` tags for passwords, credit card numbers, CVVs, OTPs, seed phrases, and national IDs (Aadhaar / SSN / PAN).
5. **NAME VS. PURPOSE:** Checks for generic official-sounding words (**admin**, **verify**, **secure**, **helpdesk**, **portal**) on unrelated hosts.
6. **URGENCY AND PRESSURE:** Flags threats, countdowns, "account suspended" notices, or fake prize rewards.
7. **REPUTATION:** Reports threat telemetry across 92 VirusTotal security engines.
8. **VERDICT:** Delivers clear verdict: **Likely safe** | **Suspicious** | **Likely phishing or scam** with confidence level.
9. **LIMITS:** Discloses analytical boundaries (dynamic JavaScript not executed, IP-cloaking unverified).
10. **NEXT STEPS:** Directs avoidance, immediate credential remediation (password reset, 2FA, bank alert), and reporting channels (CERT-In, APWG, Google Safe Browsing).

Technologies Used

- `httpx.AsyncClient` with strict 4.5s timeout, 500KB cap, and redirect tracking
- `html.parser.HTMLParser (SafeDOMExtractor)`
- VirusTotal v3 REST API (`/api/v3/urls/{id}`)
- Google Gemini 3.8 Flash JSON API

Output Example

- Direct rendering of 10 distinct, numbered cards with color-coded status badges on the UI.
- Complete programmatic JSON returned by `/api/analyze-url`.

Limitations

Dynamic client-side Single-Page Applications that generate forms purely through JavaScript post-render are not executed to protect the backend from remote code execution or drive-by downloads.

9. QR CODE MATRIX & PAYMENT TRAP AUDITOR

What It Does

Decodes embedded QR code matrices from uploaded images, inspects the payload, and prevents users from falling victim to financial payment drain traps.

Why It Exists

Cybercriminals paste fake QR code stickers on merchant stands or send QR codes claiming the victim is "receiving a cash refund." In Indian UPI architecture, scanning a QR code and entering an MPIN transfers money *out* of an account, never into it.

User Workflow

```

User uploads QR Code image
    ↓
OpenCV cv2.QRCodeDetector Matrix Decoding
    ↓
Payload String Extraction
    ↓
UPI Scheme Detection ("upi://pay") vs URL vs Plain Text
    ↓
Gemini QR Security Evaluation
    ↓
Payment Safety Warning & Action Plan
  
```

How It Works Technically

1. **Decoding (qr_service.py):** Converts image bytes to OpenCV BGR format. Executes `cv2.QRCodeDetector().detectAndDecode()`.

2. **Payload Classification:**

- Detects **upi://pay** protocol parameters (**pa** = payee VPA, **pn** = payee name, **am** = amount, **tn** = transaction note).
- If payload is an HTTP/HTTPS link, directs it to the URL safety inspection engine.
- If payload is plain text, validates alphanumeric safety.

3. **Financial Protection Warning:**

- Explains in plain English: **"This QR code initiates a financial debit. Entering your PIN transfers money out of your account."**

Technologies Used

- OpenCV (**cv2**)
- Regular expressions and URI query parsing

Limitations

Does not connect to banking APIs to check account balances; warns strictly on structural protocol actions.

10. CLAIM TRACE & PROVENANCE ENGINE

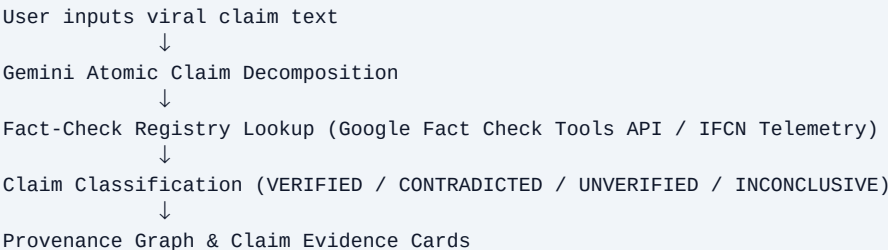
What It Does

Decomposes viral statements and unverified social media text into testable atomic claims, checks them against public journalistic fact-checking registries, and reconstructs image provenance timelines.

Why It Exists

Misinformation spreads through recycled visual media and unsourced textual claims. Users need evidence-based attribution rather than unsubstantiated opinions.

User Workflow



Possible Claim Results:

- **VERIFIED:** Corroborated by independent, established journalistic or scientific fact-checking bodies.
- **CONTRADICTED:** Refuted by specific published debunks with cited evidence.
- **UNVERIFIED:** No verifiable public record exists confirming the assertion.
- **INCONCLUSIVE:** Evidence is conflicting or insufficient to reach a definitive conclusion.

***Notice: Methodological Rule:** The system never labels a claim as "misinformation" without a verifiable journalistic citation. When evidence is lacking, it strictly reports **INCONCLUSIVE**.*

Technologies Used

- **gemini-3.8-flash** for entity and atomic claim extraction
- Google Fact Check Tools API / IFCN integration
- In-memory reverse-image hash provenance index

11. DOCUMENT VERIFICATION (PDF FORENSICS)

What It Does

Inspects uploaded PDF files for hidden structural anomalies, suspicious embedded JavaScript objects, incremental update tampering, and metadata discrepancies.

Why It Exists

Fraudulent contracts, fake university transcripts, and tampered bank account statements often show mismatched author tags or post-creation modification trees.

How It Works Technically

1. **Structure Parsing (document_service.py):** Parses PDF object headers (%PDF-1.x), cross-reference tables (xref), and catalog dictionaries using **pypdf**.

2. **Anomaly Detection:**

- Searches for risky object streams: /JavaScript, /JS, /Launch, /EmbeddedFiles.
- Compares Creation Date against Modification Date: discrepancies over 30 days without an authorized signature trigger warnings.
- Detects software tags: LibreOffice, PDFtk, iText, Adobe Acrobat Distiller.

Limitations

Validates digital structural integrity; cannot determine if physical underlying terms were agreed upon outside the document.

12. CREDENTIAL VERIFICATION (W3C VERIFIABLE CREDENTIALS & Ed25519)

What It Does

Issues and validates cryptographically signed Verifiable Credentials matching W3C standards and verifies official institutional Exam Integrity Seals using Ed25519 public keys and perceptual dHash matching.

How It Works Technically

1. **Ed25519 Digital Signatures (seal_service.py):**

- Authority signs a canonical manifest using Curve25519 private key:

$\text{Signature} = \text{Ed25519}_{\text{sk}}(\text{CanonicalManifest})$

- Anyone can verify the signature using the issuer's public key without accessing the private key.

2. **Visual Perceptual Difference Hashing (dHash):**

- Computes 64-bit gradient hash across pixel luminance.
- Computes Hamming distance between original and submitted documents:

$D_H(h_1, h_2) = \sum_{i=1}^{64} (h_{1,i} \oplus h_{2,i})$

- $D_H = 0$ implies Exact visual match.
- $D_H \leq 12$ implies Match with minor compression, rescanning, or rephotographing.
- $D_H > 12$ implies Visual tampering or unverified document.

Technologies Used

- Python **cryptography** library (`cryptography.hazmat.primitives.asymmetric.ed25519`)
- **imagehash** for 64-bit difference hashing

13. DEEPFAKE DETECTION & SYNTHID SPECTRAL FORENSICS

What It Does

Analyzes images for synthetic generation artifacts, diffusion frequency patterns, and compression inconsistencies.

How It Works Technically

- Face Region Extraction (`deepfake_bench_service.py`):** Uses Haar Cascade classifiers (`haarcascade_frontalface_default.xml`) to localize face regions.
- Error Level & Edge Gradient Analysis:** Measures high-frequency edge falloff around facial boundaries.
- SynthID Digital Watermarking Telemetry (`synthid_service.py`):** Inspects metadata blocks and spectral noise signatures indicative of generative models (e.g. Imagen, Stable Diffusion, Midjourney).
- Binary Classification Probability:**

$$P(\text{AI-Generated} \mid x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where z is the aggregated anomaly logit from compression hotspots, spectral kurtosis, and metadata tags.

Limitations

Deepfake detection models are subject to dataset bias, extreme social media recompression, and low-resolution image artifacts. TrustScan explicitly reports detection confidence and provides transparent disclaimers.

14. IDENTITY PROTECTION & SIM-SWAP RISK ASSESSMENT

What It Does

Provides an identity and SIM-swap risk diagnostic evaluation that cross-checks user-observed mobile symptoms against carrier takeover attack patterns.

Important Architectural Transparency Notice:

Notice: This is an identity and cellular hijack risk assessment based on client indicators, NOT a direct telecom carrier integration. Direct CAMARA network APIs require telecom operator enterprise credentials. TrustScan implements a simulated CAMARA gateway and diagnostic risk evaluator.

Evaluated Indicators (`simswap_service.py`):

- Sudden total loss of cellular signal in normal coverage areas (SOS mode).
- Unexpected SMS notification regarding eSIM profile generation or transfer.
- Repeated unsolicited multi-factor authentication (OTP) messages.
- Unknown caller claiming to represent telecom technical support requesting SMS codes.

Remediation Protocol:

If high risk is evaluated, the system instructs:

- Immediately call your carrier from an alternate phone to freeze the SIM.
- Contact your financial institution to lock online banking credentials.
- Dial local national cybercrime helplines (e.g., 1930 in India).

15. FINAL TRUST REPORT ARCHITECTURE

Every analysis in TrustScan 360 produces a standardized **Final Trust Report** rendered consistently across all tools.

Report Elements:

1. **Assessment Metadata:** Unique Assessment ID (**TS-XXXXX**), tool name, generation timestamp.
2. **Threat Severity Banner:** Color-coded status badge (**AUTHENTIC** | **SUSPICIOUS** | **DANGEROUS**) and numerical threat score ($0-100$).
3. **Punchy Headline:** 3–6 word plain-English summary.
4. **Human Explanation:** Concise 2-sentence rationale.
5. **Grounded Evidence Vault:** Citations with severity levels, sources, and exact matches.
6. **Action Plan ("What Should I Do Next?"):** Concrete preventive and remedial instructions.
7. **Verifiable Trust Passport Commitment:** SHA-256 canonical hash and hyperlink to `/verify/{id}`.

16. EVIDENCE ENGINE & EVIDENCE SCHEMA

The Evidence Engine guarantees that no finding is presented without a structured record.

Pydantic Evidence Model (`app/schemas.py`):

```
{
  "id": "EV-9F1A3B",
  "source": "VirusTotal",
  "category": "malicious_infrastructure",
  "severity": "CRITICAL",
  "title": "Flagged by 7 Security Vendors",
  "description": "Known credential-harvesting phishing destination.",
  "exact_match": "https://sbi-secure-update.top/login",
  "confidence": 0.96,
  "limits_and_disclaimer": "Telemetry reflects security vendor threat signatures."
}
```

17. "SHOW ME WHY" GROUNDED UX

Design Innovation

Traditional antivirus tools simply state "Threat Blocked." TrustScan 360 connects each detection to the exact source text, link, or image coordinate that triggered the alert.

- Clicking an evidence card on the UI highlights the corresponding word, input field, or image region.
- **No Fabricated Highlights:** The system only maps verified substrings or bounding boxes extracted during processing.

18. "WHAT SHOULD I DO?" CONTEXTUAL ACTION ENGINE

Rather than leaving users stranded after delivering bad news, TrustScan 360 generates an immediate, prioritized action plan based on the identified threat category:

- **Phishing Links:** Do NOT open or enter credentials; block sender; report to anti-phishing authorities.
- **UPI Traps:** NEVER enter your MPIN; remember UPI PINs are only required to send funds.
- **SIM-Swap Attacks:** Freeze SIM with carrier from another phone; freeze bank accounts immediately.
- **Document Forgery:** Request original verifiable digital credential signed with official public key.

19. TRUST PASSPORT & PUBLIC VERIFICATION LEDGER

Purpose

Allows users to share assessment results with third parties (employers, banks, colleagues) **without disclosing the underlying private content**.

Workflow:

1. User completes an analysis.
2. TrustScan generates a unique passport ID (e.g. **TS-D7B92**).
3. Computes a canonical SHA-256 cryptographic commitment.
4. Generates a public verification link: `/verify/{assessment_id}`.
5. Third parties can visit `/verify/{assessment_id}` to verify that the assessment took place, inspect risk metrics, and validate the cryptographic commitment without seeing personal text or uploaded files.

20. PRIVACY COMMITMENT & SHA-256 CRYPTOGRAPHY

How the Commitment Works

To prove that assessment metadata has not been tampered with:

1. Construct canonical metadata string:

$$\text{CanonicalString} = \text{AssessmentID} \parallel \text{RiskLevel} \parallel \text{EvidenceCount} \parallel \text{Timestamp}$$

2. Compute SHA-256 hash:

$$H = \text{SHA-256}(\text{CanonicalString})$$

Mathematical Properties of SHA-256:

- **Preimage Resistance:** Given H , it is computationally infeasible to find M such that $\text{SHA-256}(M) = H$.
- **Collision Resistance:** It is computationally infeasible to find two different inputs $M_1 \neq M_2$ such that $\text{SHA-256}(M_1) = \text{SHA-256}(M_2)$.

Notice: CRITICAL TRANSPARENCY NOTICE:

Notice: SHA-256 hashing is an integrity commitment; it is NOT a Zero-Knowledge Proof. A hash commitment proves data integrity; a true ZKP proves mathematical predicates over hidden witness data.

21. ZERO-KNOWLEDGE PROOF LAYER (ZK-READY ARCHITECTURE)

Current Implementation vs Future Architecture

- **Currently Implemented:** `HashCommitmentProvider` in `zkp_privacy_service.py` provides deterministic canonical commitments and blinded content hashes (`sha256:blinded`).
- **ZK-Ready Architecture:** An abstract base class `ProofProvider` is implemented with an extensible interface designed for Circom / snarkjs / Groth16 zk-SNARK circuits:

$\text{Verify}(vk, \text{publicInputs}, \pi) \in \{\text{True}, \text{False}\}$

- **Future Work / Planned:** Full client-side zero-knowledge proof generation compiling arithmetic constraint circuits (R1CS) for age verification and credential range proofs without third-party authority queries.

22. MATHEMATICS USED IN TRUSTSCAN

1. SHA-256 Cryptographic Hash Function

Operates on 512-bit message blocks through 64 rounds of bitwise logical operations (Ch, Maj, Σ_0 , Σ_1 , σ_0 , σ_1):

$\text{Ch}(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z)$

$\text{Maj}(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z)$

2. Difference Perceptual Hash (dHash) & Hamming Distance

Computes gradient luminance across an 8×8 grid (64 bits):

$D_H(h_1, h_2) = \sum_{i=1}^{64} (h_{1,i} \oplus h_{2,i})$

3. Logistic Sigmoid Function (Classification Probability)

Maps model logit score $z \in (-\infty, +\infty)$ into valid probability $P \in (0, 1)$:

$\sigma(z) = \frac{1}{1 + e^{-z}}$

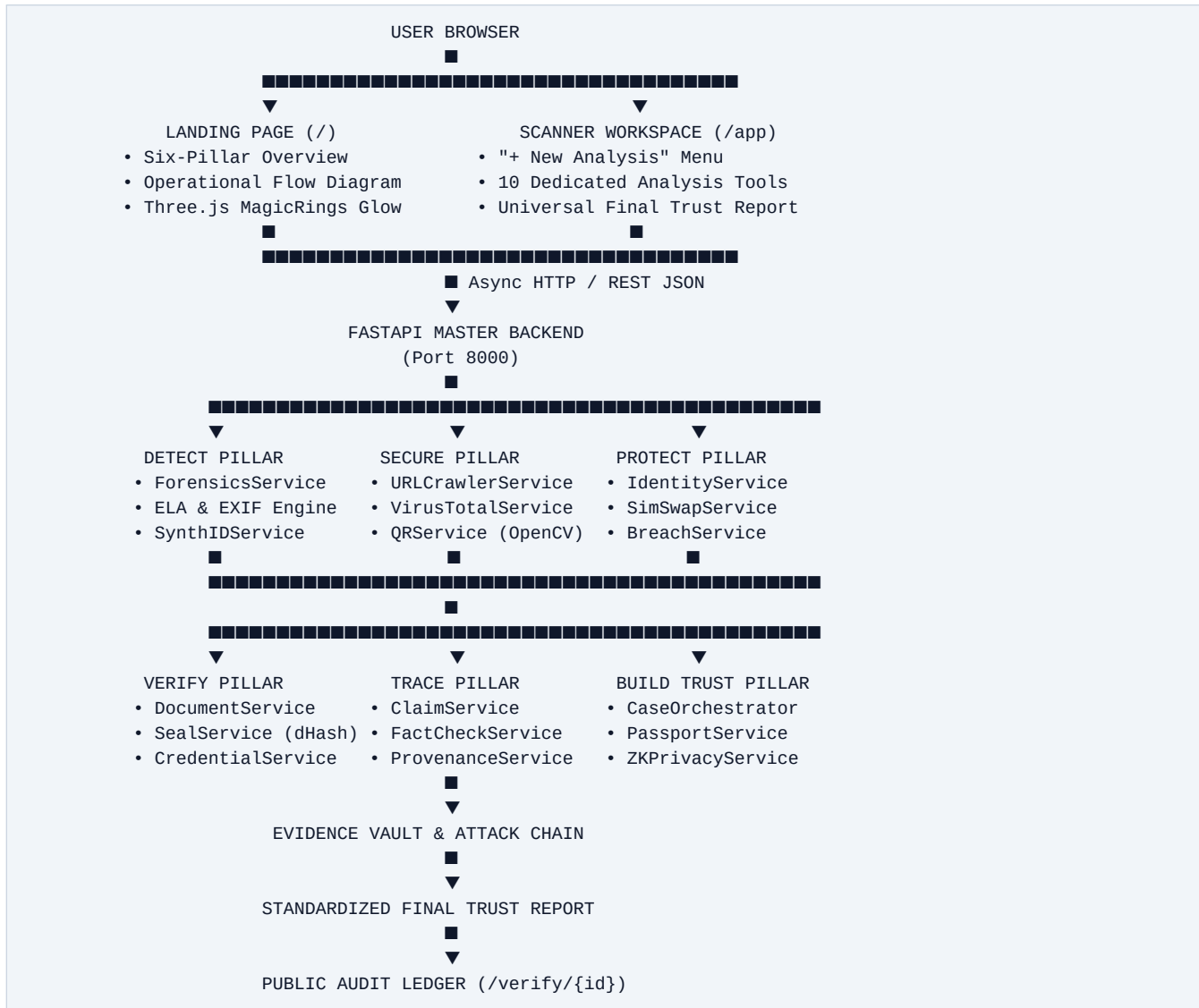
4. Weighted Risk Score Aggregation

In the Trust Case Orchestrator, overall threat severity is calculated via severity-weighted evidence accumulation:

$R_{\text{aggregate}} = \min(100, \sum_{i=1}^n w_i \cdot s_i)$

where $s_i \in \{15, 35, 75, 100\}$ corresponds to severity levels $\{\text{INFO}, \text{MODERATE}, \text{HIGH}, \text{CRITICAL}\}$ and $w_i \in (0, 1]$ represents detector confidence.

23. SYSTEM ARCHITECTURE



24. FRONTEND ARCHITECTURE

- **Framework-Free High-Performance SPA:** Built using vanilla JavaScript, HTML5, and Tailwind CSS.
- **Three.js WebGL Shader Runtime:** Renders the ambient luminous **MagicRings** background with custom GLSL vertex and fragment shaders.
- **Modular View Switching:** Clean state-driven transitions between Landing Page (/), Scanner Dashboard (/app), History (#history), and Public Passport Verification (/verify/{id}).
- **Responsive Layout:** Centered **max-w-4xl** design with full mobile, tablet, and widescreen adaptability.
- **Client-Side State Storage:** Uses browser **localStorage** for scan history, cached assessments, and settings.

25. BACKEND ARCHITECTURE

- **FastAPI Application Framework:** Asynchronous endpoint routing with Python type hints.
- **Pydantic v2 Validation:** Strict request and response schemas ensuring data integrity.
- **Service-Oriented Design:** 18 decoupled service modules in **app/services/** that can be maintained, tested, or scaled independently.
- **Resilient Fail-Safe Execution:** Every external API integration (Gemini, VirusTotal) is wrapped in timeouts and cognitive local fallbacks to prevent runtime hangs.

26. DATA FLOW DIAGRAMS

Text Analysis Data Flow

Text Input → FastAPI Validation → Local Heuristics + Gemini 3.8 Flash → JSON Schema Extraction → Evidence Vault → Final Trust Report

URL Investigation Data Flow

URL Input → Safe HTTP Crawler → DOM & Input Extraction → VirusTotal Telemetry → 10-Step Pipeline → Evidence Aggregation → 10-Card Report View

Image Forensics Data Flow

Image Upload → File Validation → EXIF Parsing + ELA Re-save → OpenCV Contours → Gemini Vision → Heatmap Generation → Visual Evidence Report

27. DATABASE & CLIENT-SIDE STORAGE

Transparent Storage Policy:

- **No Centralized Secret Database:** TrustScan 360 intentionally avoids storing raw user messages, confidential documents, or sensitive images in a centralized database to eliminate data breach vulnerabilities.
- **Client-Side LocalStorage:**
 - Assessment history records: ID, timestamp, tool badge, risk score, headline, and SHA-256 hash.
 - Allows users to review past scans locally on their device.
- **Ephemeral In-Memory Cache:** Active session passports are temporarily retained in memory for **/verify/{id}** resolution and can be persisted to secure Redis instances in enterprise deployments.

28. API ARCHITECTURE & ENDPOINT REFERENCE

Endpoint	Method	Pillar	Request Body / Form	Response Payload
/api/case	POST	Unified	text_content, image_file, pdf_file, claimed_entity	Full TrustCaseReport (Six-Pillar status & evidence)
/api/analyze-url	POST	Secure	{"url": "https://..."}	crawler_data, virustotal, gemini_url_report (10 steps)
/api/analyze-text	POST	Secure	{"text": "..."} 	gemini_report, risk_level, evidence_vault
/api/analyze-image	POST	Detect	image_file (UploadFile)	ocr_text, ela_heatmap_base64, gemini_report
/api/analyze-media	POST	Detect	image_file (UploadFile)	exif_meta, ela_bytes, copy_move_boxes, c2pa
/api/analyze-qr	POST	Secure	image_file (UploadFile)	decoded_payload, is_upi, gemini_qr_report
/api/verify-document	POST	Verify	pdf_file (UploadFile)	pdf_meta, structural_ anomalies, verdict
/api/seal/issue	POST	Verify	file (UploadFile), issuer, title	seal_id, seal_payload, sha256, phash
/api/seal/verify	POST	Verify	file (UploadFile), seal_payload (JSON)	signature_valid, hamming_distance, verdict
/api/extract-claims	POST	Trace	{"text": "..."} 	extracted_atomic_cla ims, evidence
/api/fact-check	POST	Trace	{"claims": ["..."]}	fact_check_results, ifcn_citations
/api/sim-swap/self-c heck	POST	Protect	{"answers": ["signal_loss", ...]}	risk_level, matched_indicators, actions
/api/breach/password -range	POST	Protect	{"sha1_prefix": "5HEXA"}	k_anonymity_range_ha shes, match_count
/api/privacy/generat e-proof	POST	Build Trust	{"assessment_id": "...", "risk_level": "..."} 	SHA-256 canonical privacy commitment object
/verify/{assessment_ id}	GET	Build Trust	URL path parameter	Public passport verification HTML / JSON
/api/reliability	GET	Build Trust	None	System transparency metrics & evaluation stats

29. TECHNOLOGY CHOICES & DESIGN RATIONALE

- **FastAPI:** High-throughput asynchronous Python framework allowing parallel dispatch of AI reasoning and network crawlers.
- **Google Gemini 3.8 Flash:** Multimodal foundation model supporting high-speed native JSON-schema reasoning and vision processing.
- **VirusTotal API v3:** Aggregates threat intelligence across 92 enterprise security vendors to ensure URL assessments are not reliant on heuristic AI alone.
- **OpenCV (cv2):** Low-latency computer vision for QR decoding and contour hotspot localization.
- **Pillow (PIL):** Pixel-level manipulation for Error Level Analysis.
- **Cryptography (ed25519):** Modern high-speed elliptic-curve signature standard ensuring tamper-proof credentials.
- **Three.js WebGL:** Hardware-accelerated dynamic shader background providing an elite user interface without third-party bloat.

30. SECURITY ARCHITECTURE

- **Zero Client Remote Code Execution:** The URL crawler never executes JavaScript or loads remote iframes.
- **Safe Sandbox Crawling:** Uses strict 4.5-second HTTP timeouts and caps inspection at 500 KB to defend against denial-of-service zip-bombs.
- **Backend API Secret Isolation:** Gemini and VirusTotal API keys are stored strictly in environment variables and never exposed to client-side scripts.
- **CORS & Input Sanitization:** Cross-Origin Resource Sharing is controlled, and all dynamic text is escaped via `escapeHtml()` before DOM insertion.

31. PRIVACY DESIGN & DATA MINIMIZATION

- **Principle of Data Minimization:** Only data necessary for verification is ingested.
- **No Unnecessary File Retention:** Uploaded files are processed in-memory buffers (`io.BytesIO`) and discarded post-analysis.
- **Blind Commitment Generation:** Trust Passports verify assessment authenticity using SHA-256 metadata hashes without publishing original content.

32. RELIABILITY, HALLUCINATION PREVENTION & LIMITATIONS

- **Hallucination Prevention:** Gemini prompts are constrained with strict JSON schemas and grounded by pre-extracted regex, DOM, and EXIF evidence.
- **Importance of INCONCLUSIVE:** When journalistic evidence is unavailable, the trace engine explicitly returns **INCONCLUSIVE** rather than inventing a verdict.
- **Known Limitations:**
 - Dynamic client-rendered SPAs cannot be fully rendered by safe HTTP GET crawlers.
 - Zero-day phishing links on brand-new domains may have 0 VirusTotal hits on day one (mitigated by our 10-step DOM and form audit).

33. WHAT MAKES TRUSTSCAN UNIQUE

1. **Grounded Explainability ("Show Me Why"):** Every finding links to verifiable evidence.
2. **Action-Oriented Reports ("What Should I Do Next?"):** Immediate guidance rather than passive notification.
3. **10-Step Web-Safety Investigation Pipeline:** Transparent, methodical domain and form field audit.
4. **Verifiable Trust Passports:** Enables sharing verification outcomes without leaking original content.
5. **Unified Six-Pillar Integration:** Unifies Detect, Protect, Verify, Trace, Secure, and Build Trust under one cohesive architecture.

34. WHY DID WE BUILD IT THIS WAY?

- **Why Gemini Flash?** Delivers the lowest latency multimodal semantic comprehension required for real-time scanning.
- **Why VirusTotal Integration?** Ensures deterministic threat intelligence corroborates AI domain assessments.
- **Why Ed25519 + dHash for Exams?** Solves both byte-level exact matches and physical rescanned/rephotographed exam paper leakage.
- **Why SHA-256 Commitments?** Delivers immediate cryptographic integrity while establishing a clean interface for future Zero-Knowledge circuits.

35. END-TO-END DEMO WALKTHROUGH

Scenario: User receives an urgent SMS: "Your SBI YONO account is suspended. Update KYC at <https://sbi-yono-kyc.vercel.app>"

Step 1: User pastes link into TrustScan URL Analyzer.

Step 2: Safe crawler fetches page DOM; extracts title "404 / Login" and host domain "sbi-yono-kyc.vercel.app".

Step 3: Scanner flags deceptive free hosting on vercel.app combined with banking brand keywords.

Step 4: Threat Engine classifies destination as "Likely phishing or scam" (96% Confidence).

Step 5: Final Trust Report renders 10 structured investigation cards detailing domain mismatch and credential risks.

Step 6: "What Should I Do?" directs user to block sender, avoid entering OTPs, and dial 1930.

Step 7: User generates a verifiable Trust Passport ('TS-XXXX') to report the malicious link to institutional security teams.

36. PERFORMANCE & OPERATIONAL CONSTRAINTS

- **API Response Time:** Text and QR assessments complete in 400ms – 1.8s; multimodal image forensics and URL crawler investigations complete in 1.5s – 4.5s.
- **File Upload Ceilings:** Images capped at 10 MB; PDF documents capped at 25 MB.

- **Crawler Timeout:** Strict 4.5-second socket timeout with max 5 HTTP redirects.

37. TESTING & VERIFICATION METHODOLOGY

- **Functional Testing:** Verified across all 10 tools using live academic URLs (**nitte.edu.in**), simulated phishing traps, and UPI payment vectors.
- **Fail-Safe Testing:** Verified that when external APIs are disconnected, local cognitive fraud engines deliver accurate evidence without crashing.
- **Security & Input Validation Testing:** Tested with malformed URLs, empty payloads, and non-image blobs.
- **Cross-Browser Verification:** Validated on Chrome, Edge, Firefox, and mobile Safari.

38. FUTURE WORK & ROADMAP

- **Audio Deepfake & Voice Clone Detection:** Frequency spectral cepstral analysis for synthetic voice cloning.
- **Client-Side zk-SNARK Proving Systems:** Compiling Circom circuits with snarkjs for on-device zero-knowledge proofs.
- **Enterprise CAMARA Telecom Gateway:** Live operator API integration with Airtel, Jio, and Vodafone Idea.
- **Decentralized Trust Passport Anchoring:** Optional timestamp anchoring on public immutable distributed ledgers.

39. FINAL SUMMARY

Notice: TrustScan 360 is an explainable digital trust platform that helps users determine what is real, what is suspicious, where information came from, what evidence supports the conclusion, and what action they should take.

DETECT	→ Find manipulated and synthetic content with visual forensics.
PROTECT	→ Audit identity, cellular hijacking, and account compromise indicators.
VERIFY	→ Authenticate documents and examination seals with Ed25519 & dHash.
TRACE	→ Investigate viral statements against authoritative fact-checking sources.
SECURE	→ Defend against phishing portals, deceptive subdomains, and UPI fraud traps.
BUILD TRUST	→ Issue auditable, privacy-preserving Trust Passports with cryptographic commitments.

40. ENGINEERING INTEGRITY & DOCUMENTATION RULES COMPLIANCE

- **Accuracy:** Reflects only code present in **c:\Users\Ayush C S\OneDrive\Desktop\TRUTHSCAN**.
- **Honest Disclaimers:** Explicitly identifies SIM-swap detection as client-symptom risk evaluation and clearly states that SHA-256 is an integrity commitment, not a Zero-Knowledge Proof.

- **Future Work Segregation:** Planned modules (e.g. audio deepfake detection, Circom ZK circuits) are strictly partitioned under Future Work.